

Week 8: Combinatorial Foundation

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※ Multiplication Principle

A snack shop offers: 3 kinds of sandwiches and 2 kinds of drinks. You choose one sandwich and one drink. How many different meals can you make?

Let the sandwiches be S1, S2, and S3 while the drinks are D1 and D2. Listing them out we have:

S1	D1
S2	D1
S3	D1
S1	D2
S2	D2
S3	D2

So, there are 6 ways. But, if 3 and 2 are replaced with larger numbers or there are more options, this may be tedious.

Fortunately, the **Multiplication Principle** states that when a process happens in stages, multiply the number of choices at each stage. In the above example, the sandwiches and drinks are independent and happen in separate stages. Whatever sandwich we choose is completely irrelevant from whatever drink we choose. So, we can multiply the number of options: $3 \times 2 = 6$.

For example, suppose you have:

- 4 shirts
- 3 pairs of pants
- 2 pairs of shoes

The number of different outfits is

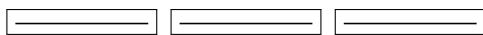
$$\underbrace{4}_{\text{shirts}} \times \underbrace{3}_{\text{pants}} \times \underbrace{2}_{\text{shoes}} = 24.$$

Each factor represents one decision.

※ Factorial

Ana, Ben, and Carlos are standing in a line for a photograph. How many ways are there to rearrange them?

Consider three seats:



There are 3 choices for the first position. Once that person has been chosen, there are 2 choices for the second position. There is then only 1 person left for the final position.

$$3 \times 2 \times 1 = 6.$$

Although this might feel weird since whatever seat the first person sits in does impact the possibilities the second person can sit in, we can still apply the multiplication principle because the number of possibilities the second person can sit in remains constant at 2 choices no matter what.

Now suppose four students are standing in a line. There are:

- 4 choices for the first position,
- 3 choices for the second position,
- 2 choices for the third position, and
- 1 choice for the last position.

Therefore, $4 \times 3 \times 2 \times 1 = 24$.

There is actually a cool tool called **factorials** that solve these problems immediately. The product

$$n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$$

is called n factorial and is written as $n!$. For example,

$$3! = 3 \times 2 \times 1 = 6,$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120,$$

I recommend memorizing up to $10!$.

※ Permutations

Five students compete in a contest. One student will receive a gold medal, one will receive a silver medal, and one will receive a bronze medal. How many different medal results are possible?

There are 5 choices for gold. After choosing the gold medalist, there are 4 choices for silver. Then there are 3 choices for bronze.

$$5 \times 4 \times 3 = 60.$$

The order matters because the medals are different. Ana winning gold and Ben winning silver is not the same result as Ben winning gold and Ana winning silver.

A selection in which order or position matters is called a permutation.

The number of ways to select and arrange r objects from n distinct objects is

$$P(n, r) = \frac{n!}{(n-r)!}$$

For the medal problem,

$$P(5, 3) = \frac{5!}{2!} = 60.$$

It is often better to write the factors and explain what they represent instead of immediately using the formula.

※ Combinations

Five students are available. Two students will be chosen to carry a banner together.

We might begin by calculating

$$5 \times 4 = 20.$$

However, this counts each pair twice. For example, choosing Ana and then Ben gives the same team as choosing Ben and then Ana.

AB and BA

represent the same pair. Therefore, we divide by 2: $\frac{5 \times 4}{2} = 10$. A selection in which order does not matter is called a combination.

This one has a nicer symbol! The number of ways to choose r objects from n distinct objects when order does not matter is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

The expression $\binom{n}{r}$ reads as “ n choose r ”.

※ Challenge Examples

In general, you will rarely see problems like the above in competitions. Those are simply too easy. Most competitions will instead put a spin on these problems. They will still use the same idea, but in a unique way. Here, I document a few examples of the “spin” problems. But, you will encounter more in the practice problems.

Six students stand in a line. Lia and Max must stand next to each other. How many lineups are possible?

Treat Lia and Max as one block. We now arrange:

- the Lia-Max block,
- and the four other students.

That gives 5 objects. Just arranging these four blocks yields $5! = 120$ ways. Within their block, Lia and Max can appear as LM or ML, giving 2 possible orders. Therefore,

$$2 \times 5! = 240.$$

Four-letter codes are made using different letters chosen from

A, B, C, D, E, F .

How many codes contain the letter A ?

There are two ways you can do this:

Way #1. There are 4 ways to place the A in some position. Then, we will place B, C, D, E , and F into the remaining three positions. This comes with

$$P(5, 3) = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 60$$

total ways. Multiplying yields $4 \times 60 = \boxed{240}$.

Way #2. Ignore the condition of having an A . Then, there are:

$$P(6, 4) = \frac{6!}{(6-4)!} = \frac{6!}{2!} = 360$$

ways. We count the complement: suppose we don't have an A . There are

$$P(5, 4) = \frac{5!}{(5-4)!} = \frac{5!}{1!} = 120$$

total ways. Subtracting yields $360 - 120 = \boxed{240}$.

※ Practice Problems

1. What is the value of $\frac{7!}{5!}$?

2. How many three-digit numbers can be formed using the digits

0, 1, 2, 3, 4, 5

if no digit may repeat? (Recall, a three-digit number cannot start with a 0.)

3. A three-person team is selected from seven students. Two of the students are Ana and Ben. How many teams contain exactly one of Ana or Ben?

4. Oscar has 7 books, and he wants to give 3 books to Chelsea. In how many ways can he choose which books to give her?

5. How many arrangements of the letters in the word CUBES have the two vowels next to each other?

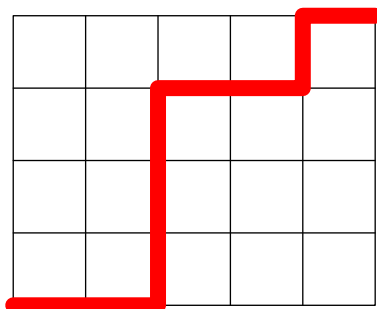
6. How many positive integer divisors does 72 have? How about 108? How about 120?

7. How many different ordered three-letter permutations can be formed using three of the seven letters in the word **ALGEBRA**?

8. The Jolly Dodger Cafe menu includes 10 entrees, 4 sides, 6 appetizers and some number of desserts. The meal deal includes exactly 1 entree and 1 side, and either 1 appetizer or 1 dessert. If there are 440 different ways to order the meal deal, how many different desserts are on the menu?

9. How many ways can you put 19 identical balls into 6 bins, so that no bin is empty?

10. How many ways are there for an ant to travel from the bottom-left to the top-right such that an ant either moves one unit to the right or one unit up? One example is shown below.



11. How many permutations of the first 9 positive integers satisfy both of these properties?

- The leftmost even number is 6.
- The rightmost odd number is 9.